

P·E·D·W Advanced

One hidden layer, real weight matrices - same four moves - the ANSWER KEY

THE SETUP - DAY D13 OF THE GOLF LOGBOOK

D13 said: hot, dry, calm ...and they PLAYED! (so this time the target is 1)

layer_0 =

Temp	Humidity	Wind
1	0	0

target y =

Play?
1

The weight matrices (random start, seed(1), rounded to 2 decimals - your hand math will check exactly):

weights_0_1 =

	node1	node2	node3	node4
Temp	-0.17	0.44	-1.00	-0.40
Humidity	-0.71	-0.82	-0.63	-0.31
Wind	-0.21	0.08	-0.16	0.37

weights_1_2 =

node1	-0.59
node2	0.76
node3	-0.95
node4	0.34

$\alpha = 0.1$

(learning rate) · $3 \times 4 = 12$ weights into the hidden layer, 4 more to the output: 16 total

TWO TINY HANDCRAFTED FUNCTIONS - THIS IS ALL THE MAGIC

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def relu(x):
    return (x > 0) * x

def relu2deriv(output):
    return output > 0
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relu: a negative number gets recoded to 0.
(positive numbers pass straight through)

relu2deriv: THE MASK. 1 if the node fired,
0 if it slept. Sleeping nodes learn nothing.

Forward: $\text{layer}_1 = \text{relu}(\text{layer}_0 \cdot W_{01})$ then $\text{prediction} = \text{layer}_1 \cdot W_{12}$ (no relu at the output!)

Backward: the delta flows back through the SAME weights - and relu2deriv silences the nodes that never fired.

THE FLIGHT PLAN (same P·E·D·W, just with matrices)

p.2 FORWARD: layer_1 before relu $\rightarrow$ relu $\rightarrow$ prediction then E and D

p.3 BACKWARD: update weights_1_2, build layer_1_delta (the mask!), update weights_0_1

FORWARD PASS + E + D · ANSWER KEY

P - step 1: $\text{layer_1 (before relu)} = \text{layer_0} \cdot W_{01}$

One dot product per hidden node - walk the columns of W_{01} with $x = [1, 0, 0]$:

node1: $(1 \times -0.17) + (0 \times -0.71) + (0 \times -0.21) = -0.17$

node2: $(1 \times 0.44) + (0 \times -0.82) + (0 \times 0.08) = 0.44$

node3: $(1 \times -1.00) + (0 \times -0.63) + (0 \times -0.16) = -1.00$

node4: $(1 \times -0.40) + (0 \times -0.31) + (0 \times 0.37) = -0.40$

	node1	node2	node3	node4
layer_1 (pre-relu) =	-0.17	0.44	-1.00	-0.40

P - step 2: $\text{layer_1} = \text{relu}(\text{layer_1})$

Negative? Recode to 0. Positive? Pass through.

	node1	node2	node3	node4
layer_1 =	0	0.44	0	0

← only node2 fired!

P - step 3: $\text{prediction} = \text{layer_1} \cdot W_{12}$ then E and D

P $(0 \times -0.59) + (0.44 \times 0.76) + (0 \times -0.95) + (0 \times 0.34) = 0.3344$

E $(1 - 0.3344)^2 = 0.443$

D $0.3344 - 1 = -0.6656$

D is NEGATIVE: we UNDERestimated (they played and we said 0.33). The messenger weights must get BIGGER.

BACKWARD PASS - THE W IN P·E·D·W · ANSWER KEY

W (part a): $\text{weight_delta_1_2} = \text{layer_1} \times D$ then update W12

Each hidden node's activation times the delta. Keep 4 decimals.

	weight_delta_1_2		new W12	
node1:	$0 \times -0.6656 =$	0	$-0.59 - 0.1 \times 0 =$	-0.59
node2:	$0.44 \times -0.6656 =$	-0.2929	$0.76 - 0.1 \times (-0.2929) =$	0.7893
node3:	$0 \times -0.6656 =$	0	$-0.95 - 0.1 \times 0 =$	-0.95
node4:	$0 \times -0.6656 =$	0	$0.34 - 0.1 \times 0 =$	0.34

Only node2's weight moved: $0.76 \rightarrow 0.7893$. It UNDERdelivered, so its volume gets turned UP.

W (part b): $\text{layer_1_delta} = D \times W12$...then APPLY THE MASK

Send the miss backward through the same weights: $D \times$ each W12 value. Keep 4 decimals.

	node1	node2	node3	node4
before the mask =	0.3927	-0.5059	0.6323	-0.2263

$\times \text{relu2deriv}(\text{layer_1}) = [0, 1, 0, 0] \rightarrow$

layer_1_delta =	0	-0.5059	0	0
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← the mask kills the sleepers

W (part c): $\text{weight_delta_0_1} = \text{layer_0}^T \times \text{layer_1_delta}$ then update W01

Outer product: each INPUT row times layer_1_delta. Inputs Humidity and Wind were 0 - whole rows of zeros!

	node1	node2	node3	node4		node1	node2	node3	node4
Temp	0	-0.5059	0	0	new W01 =	-0.17	0.4906	-1.00	-0.40
Hum	0	0	0	0		-0.71	-0.82	-0.63	-0.31
Wind	0	0	0	0		-0.21	0.08	-0.16	0.37

Only (Temp $\rightarrow$ node2) moves: $0.44 - 0.1 \times (-0.5059) = 0.4906$. 15 of 16 weights slept through this round!

THE MORAL: information only flows - forward OR backward - through nodes that FIRE.

Temp built node2, node2 built the prediction, so the credit (and the blame) lands on exactly that path.